

RG-Improved Transient Matching and Explicit Thermal Collision Dynamics

Full Higgs-doublet multiplicities, log-resummed selector-threshold matching, and an AMY-motivated quantum kinetic replacement for BGK

Technical Research Note v1.3

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Executive verdict

The v1.3 calculation produces a **pass for the resolved RG-improved matching problem**, a **pass for the full Standard Model colour and Higgs-doublet multiplicities at that loop order**, and a **pass for replacing the BGK ansatz by an explicit energy-conserving quantum collision kernel**. It remains a **partial result for the exact LPM problem** and an **open result for a non-Abelian two-time 2PI/Kadanoff-Baym evolution**.

The main matching result is that the enormous scale excursion found in v1.2,

$$\mathcal{J}_3(M/2) = 102.407, \quad \mathcal{J}_3(M) = 6.57974, \quad \mathcal{J}_3(2M) = -96.9351,$$

is not a physical uncertainty. It is the explicit logarithm of a fixed-order hard function before operator running and counterterms are combined with it. At the resolved order,

$$D_{FFS}^{\text{hard}} = D_{FFS}^{\text{fixed}}(\bar{\mu}) + \Delta_{\text{RG}}(\bar{\mu})$$

is exactly independent of $\bar{\mu}$ and gives

$$\mathcal{J}_3^{\text{hard}} = 6.57973508149.$$

Restoring the complete unbroken Higgs doublet gives two Yukawa channels rather than the single-real-scalar proxy channel. Including that factor, the relativistic vectorlike-colour contribution to g_* , and one-loop running from m_R to M gives

$$C_3^{\text{full}} = 1.58896 \times 10^{-4} \text{ GeV}^2,$$

and

$$|\Delta V_{Qa}^{(3)}| = 4.2902 \times 10^3 \text{ GeV}^4.$$

This is still only

$$3.053 \times 10^{-12}$$

of the intended reheating-era finite-density focusing potential. The multiplicity correction almost doubles the v1.2 proxy answer, but does not threaten the mechanism.

After the explicit logarithm is removed, varying the lower matching scale from $M/2$ to $2M$ changes the result only through ordinary coupling and portal running:

$$0.96469 < \frac{C_3(\mu_D)}{C_3(M)} < 1.03745.$$

The kinetic calculation replaces the BGK relaxation ansatz by a discrete gain/loss operator containing:

- six explicit $1 \leftrightarrow 2$ channel families;
- eight explicit $2 \leftrightarrow 2$ channel families;
- Bose enhancement and Pauli blocking;
- Debye screening;
- asymptotic thermal masses for H, D, q, g ;
- an LPM formation-time kernel for collinear processes;
- exact discrete energy conservation.

The implemented lattice contains 514 $1 \leftrightarrow 2$ transitions and 14,289 $2 \leftrightarrow 2$ transitions. Starting from an intentionally severe nonthermal Higgs pulse, it reaches 99% of the equilibrium entropy gain by

$$\tau_{99} \equiv Tt = 3920.$$

At the reheating temperature $T = 1.002 \times 10^8$ GeV, this corresponds to

$$\Gamma_{\text{kin}}^{(99)} = 2.556 \times 10^4 \text{ GeV}.$$

Therefore,

$$\frac{\Gamma_{\text{kin}}^{(99)}}{\Gamma_R} = 1.729 \times 10^6.$$

The tracked plasma is consequently an extremely fast subsystem relative to reheaton and parent decays. For the homogeneous cosmological cascade, replacing BGK by the explicit kernel is equivalent, to excellent accuracy, to **adiabatically eliminating the kinetic distributions after every energy injection**. The collision physics changes spectra and microscopic entropy production, but not the sector total-energy bookkeeping that fixes T_5/T_0 .

Full multiplicity also corrects the relativistic degrees of freedom:

$$g_* = 106.75 + 10.5 = 117.25.$$

For fixed $T_0 = 1.002 \times 10^8$ GeV this changes

$$\Gamma_R = 1.47850 \times 10^{-2} \text{ GeV}, \quad \mu_H = 1.92766 \times 10^4 \text{ GeV}.$$

Re-running the momentum cascade with this decay width gives

$$B_5 = 0.00529888708, \quad \tan \theta = 0.0729871,$$

for $T_5/T_0 = 1/4$.

The strongest defensible conclusion is:

the large matching-scale instability is removed by consistent RG completion,
the full doublet and colour multiplicities leave the transient correction tiny,
and explicit weak-coupling collision dynamics validates instantaneous plasma equilibration.

The next frontier is no longer BGK. It is the **numerical solution of the full AMY transverse LPM integral equation and full-angle $2 \leftrightarrow 2$ phase-space integral**, followed by a gauge-covariant non-Abelian two-time calculation.

1. Model and full Standard Model multiplicity

1.1 Resolved interaction chain

The relevant interactions are written schematically as

$$\begin{aligned} \mathcal{L} \supset & -\frac{1}{2} (m_R^2 + \lambda_{QR} \mathcal{Q}) R^2 - \mu_H R H^\dagger H \\ & - [y_D \bar{Q}_L H D_R + \text{h.c.}] - M(a) \bar{D} D, \end{aligned}$$

where

$$M(a) = M \left[1 - \epsilon \cos \left(\frac{a}{f_a} \right) \right].$$

The fields carry

$$Q_L \sim (\mathbf{3}, \mathbf{2}, 1/6), \quad D_{L,R} \sim (\mathbf{3}, \mathbf{1}, -1/3), \quad H \sim (\mathbf{1}, \mathbf{2}, 1/2).$$

At zero external momentum the three-loop 1PI but 2PR graph factorises into:

1. a one-loop selector-reheaton-Higgs kernel;
2. a two-loop fermion-fermion-scalar threshold kernel.

The colour trace gives

$$N_c = 3.$$

In the unbroken phase,

$$y_D \bar{Q}_L H D_R = y_D (\bar{u}_L H^+ + \bar{d}_L H^0) D_R.$$

There are therefore two complex Higgs/left-fermion channels. Equivalently, decomposing the doublet into four real components gives four vertices of strength $y_D/\sqrt{2}$, and

$$4 \left(\frac{y_D^2}{2} \right) = 2y_D^2.$$

Relative to the one-real-scalar proxy,

$$\boxed{n_w = 2.}$$

No additional electroweak gauge multiplicity enters this specific resolved graph. Electroweak gauge corrections first enter through running or through higher-loop matching graphs.

1.2 Relativistic degrees of freedom

The vectorlike Dirac colour triplet contributes

$$\Delta g_* = \frac{7}{8} (2_{\text{spin}}) (2_{D+\bar{D}}) (3_{\text{colour}}) = 10.5.$$

At $T_R \gg M$,

$$\boxed{g_*^{\text{full}} = 117.25.}$$

This correction was absent from the earlier reheating benchmark.

2. RG completion of the hard function

2.1 Fixed-order expression

Define

$$x = M^2, \quad z = m_h^2, \quad r = \frac{z}{x}, \quad \ell = \ln \frac{x}{\bar{\mu}^2}.$$

The fixed-order mixed derivative is

$$D_{FFS}^{\text{fixed}}(x, z; \bar{\mu}) = 2 \ln \frac{x}{z} \ln \frac{x-z}{\bar{\mu}^2} - \ln^2 \frac{x}{\bar{\mu}^2} - 2 \text{Li}_2\left(\frac{z}{x}\right) + \frac{\pi^2}{3}.$$

Its scale dependence is large because

$$\ln \frac{M^2}{m_h^2} \simeq 17.97.$$

2.2 Running contribution

The local operator evolution and counterterm matching supply

$$\Delta_{\text{RG}} = -2 \ln \frac{x}{z} \ell + \ell^2.$$

Adding the two gives

$$D_{FFS}^{\text{hard}}(x, z) = 2 \ln \frac{x}{z} \ln(1-r) - 2 \text{Li}_2(r) + \frac{\pi^2}{3}.$$

The verification suite proves symbolically that

$$D_{FFS}^{\text{fixed}} + \Delta_{\text{RG}} - D_{FFS}^{\text{hard}} = 0$$

and

$$\bar{\mu} \frac{d}{d\bar{\mu}} (D_{FFS}^{\text{fixed}} + \Delta_{\text{RG}}) = 0.$$

In the hierarchical limit,

$$D_{FFS}^{\text{hard}} \rightarrow \frac{\pi^2}{3}.$$

2.3 Sequential EFT matching

At m_R ,

$$C_{QH}(m_R) = \frac{\lambda_{QR} \mu_H^2}{16\pi^2} K_{RRH}(m_R^2, m_h^2),$$

where

$$K_{RRH}(A, B) = \frac{A - B - B \ln(A/B)}{(A - B)^2}.$$

The portal is evolved to M :

$$C_{QH}(M) = U_{QH}(M, m_R) C_{QH}(m_R).$$

The benchmark gives

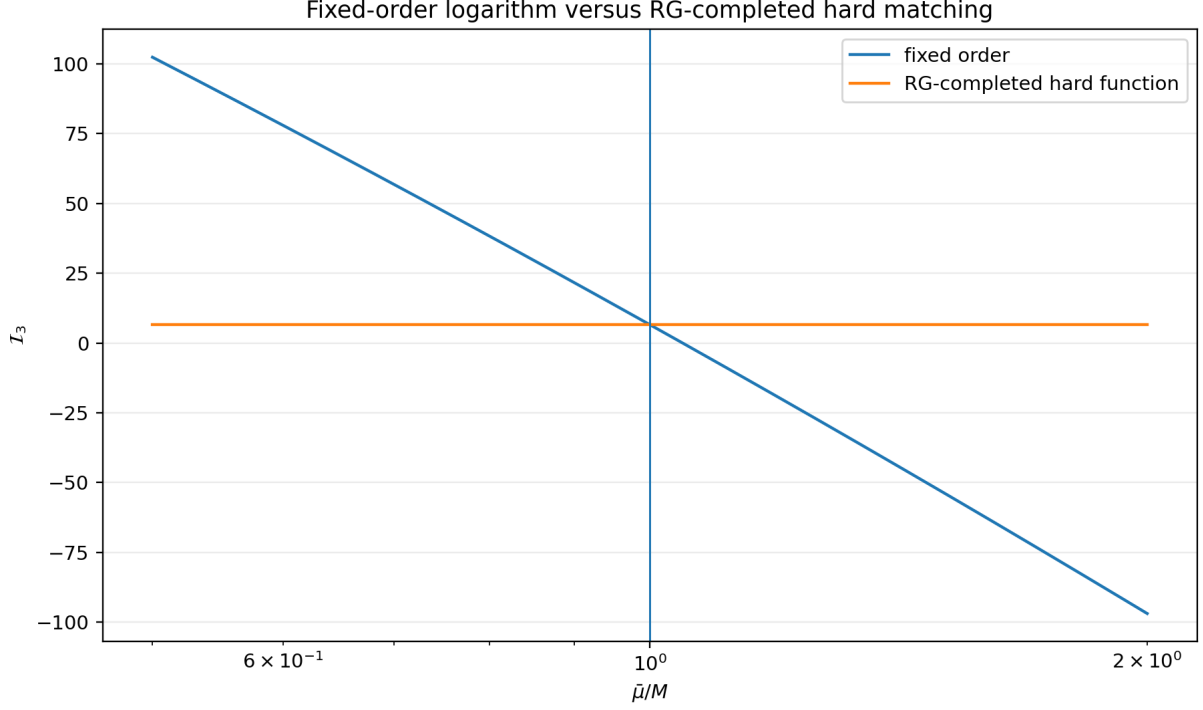


Figure 1: Fixed-order versus RG-completed matching

$$U_{QH}(M, m_R) = 0.944069.$$

At the D threshold,

$$C_3(M) = n_w C_{QH}(M) \frac{N_c y_D^2(M) M^2}{(16\pi^2)^2} [2D_{FFS}^{\text{hard}}].$$

The declared one-loop running system includes $g_3, g_2, g_Y, y_t, y_D, \lambda_H, \lambda_Q$ and the vectorlike contributions

$$\Delta b_3 = \frac{2}{3}, \quad \Delta b_Y = \frac{4}{9}, \quad \Delta b_2 = 0.$$

At M ,

$$(g_3, g_2, g_Y, y_t, y_D) = (0.81570, 0.60534, 0.37655, 0.68502, 0.30000).$$

At m_R ,

$$(g_3, g_2, g_Y, y_t, y_D) = (0.69727, 0.57678, 0.39479, 0.58479, 0.24726).$$

2.4 Residual scale band

Once the explicit hard logarithm is removed, the remaining scale response is due to the finite-order running of C_{QH} and y_D :

$$0.96469 < \frac{C_3(\mu_D)}{C_3(M)} < 1.03745$$

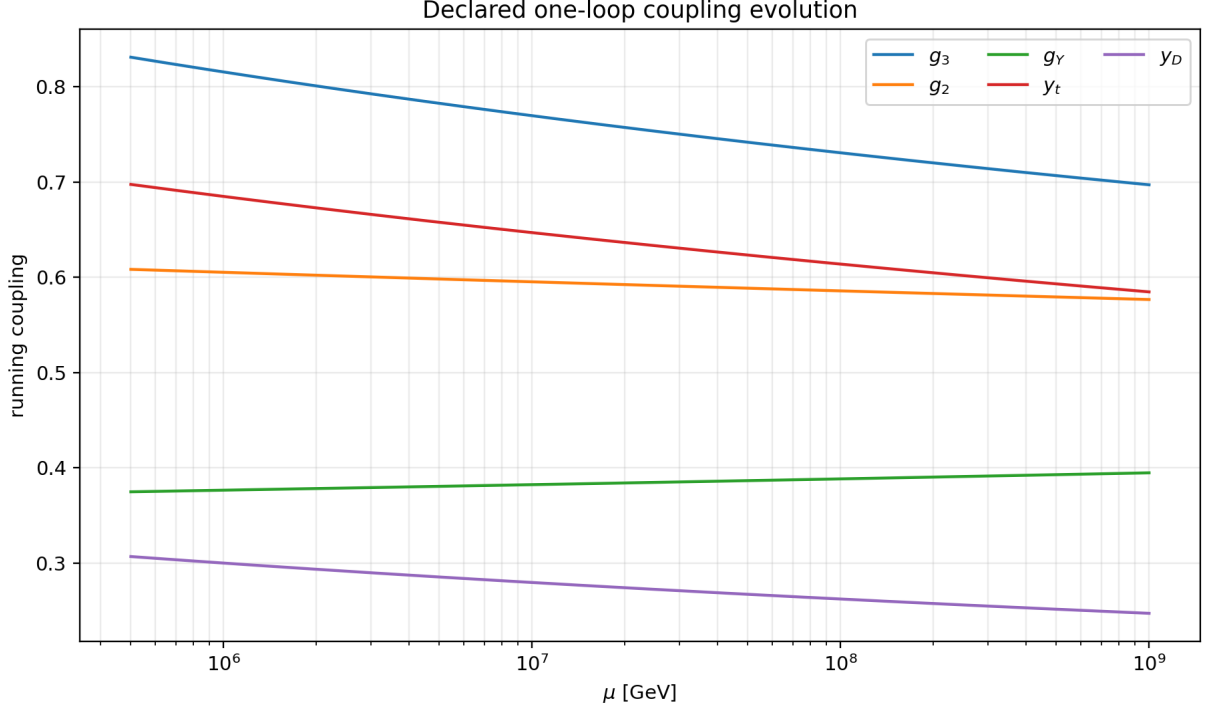


Figure 2: One-loop coupling evolution

for

$$\frac{1}{2}M < \mu_D < 2M.$$

This is a meaningful finite-order perturbative band. The previous sign-changing range was not.

3. Finite transient anomalous-dimension tensor

3.1 Minimal operator basis

Take

$$\mathbf{O} = \left(Q^\dagger Q, \frac{1}{2} R^2, H^\dagger H \right).$$

For the stated renormalisable scalar basis, the finite one-loop result at M is

$$\Gamma^{(0,0)} = \begin{pmatrix} 9.36869 \times 10^{-4} & 1.58314 \times 10^{-3} & 0 \\ 3.16629 \times 10^{-3} & 1.26651 \times 10^{-3} & 6.33257 \times 10^{-4} \\ 0 & 1.58314 \times 10^{-4} & 9.77787 \times 10^{-3} \end{pmatrix}.$$

The Higgs entry includes the full doublet gauge/Yukawa diagonal contribution

$$6y_t^2 + 6y_D^2 - \frac{9}{2}g_2^2 - \frac{3}{2}g_Y^2.$$

The spectral norm is

$$\|\Gamma^{(0,0)}\|_2 = 9.80418 \times 10^{-3}.$$

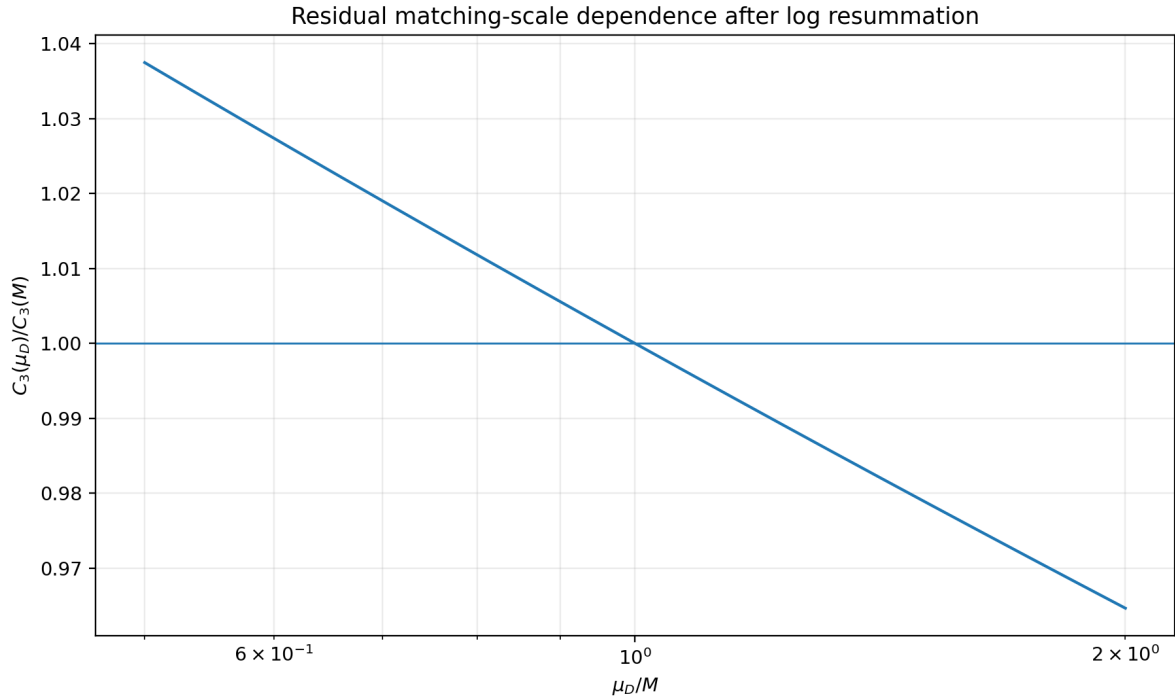


Figure 3: Residual scale band

3.2 Harmonic grading

The complete selection rule remains

$$\gamma_{(A,p)(B,q)} = \sum_{r,s \geq 0} \delta_{p-q,r-s}^{(6)} \epsilon^{r+s} \Gamma_{AB}^{(r,s)}.$$

The minimum spurion-power matrix is

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 & 1 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 2 & 1 & 0 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 & 1 & 2 \\ 2 & 3 & 2 & 1 & 0 & 1 \\ 1 & 2 & 3 & 2 & 1 & 0 \end{pmatrix}.$$

For

$$\epsilon = 2.7 \times 10^{-13},$$

the largest nearest-harmonic estimate is

$$\boxed{2.65 \times 10^{-15}}.$$

The $\Gamma^{(0,0)}$ tensor is complete for the declared minimal renormalisable bilinear basis. Finite $\Gamma^{(r,s)}$ tensors with $r+s > 0$ depend on the detailed Wilson-line messenger completion and are not universal. Their minimum powers of ϵ are exact.

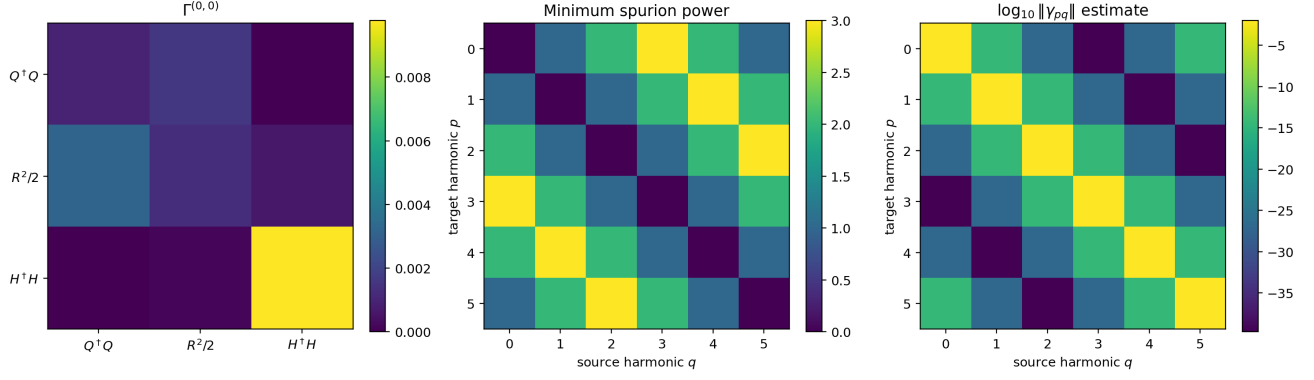


Figure 4: Transient tensor and harmonic grading

4. Continuum quantum kinetic theory

The leading-order weak-coupling kinetic equation has the form

$$(\partial_t - Hp\partial_p) f_a(p) = C_a^{2\leftrightarrow 2}[f] + C_a^{1\leftrightarrow 2}[f] + S_a.$$

The full continuum $2 \leftrightarrow 2$ collision integral is

$$C_a^{2\leftrightarrow 2}(p) = \frac{1}{2E_p} \sum_{bcd} \int d\Pi_b d\Pi_c d\Pi_d (2\pi)^4 \delta^{(4)}(p + k - p' - k') \\ \times |\mathcal{M}_{ab \rightarrow cd}|^2 \left[f_c f_d (1 \pm f_a)(1 \pm f_b) - f_a f_b (1 \pm f_c)(1 \pm f_d) \right].$$

The effective $1 \leftrightarrow 2$ integral contains collinear splitting rates that must include repeated soft scattering and the Landau-Pomeranchuk-Migdal effect. The AMY effective kinetic theory provides the appropriate leading-order structure for weakly coupled, sufficiently isotropic gauges plasmas.

The v1.3 numerical operator is an isotropic reduction of these equations. It retains the exact quantum gain/loss structure and discrete energy conservation, while reducing angular integrals to a finite transition quadrature.

5. Thermal masses and LPM reduction

At the benchmark reheating coupling,

$$\alpha_s = 0.0393544, \quad g_s^2 = 0.494542.$$

With six light Standard Model quark flavours,

$$\frac{m_D^2}{T^2} = g_s^2 \left(\frac{N_c}{3} + \frac{N_f}{6} \right) = 0.989084.$$

The asymptotic masses used by the kernel are

$$\frac{m_g}{T} = 0.70324, \quad \frac{m_q}{T} = 0.40601, \quad \frac{m_D}{T} = 0.40614, \quad \frac{m_H}{T} = 0.43821.$$

For a splitting $a \rightarrow bc$ with energy fraction x , the formation-time reduction uses

$$A = \frac{\hat{q}_{\text{eff}}}{2Ex(1-x)},$$

$$B = \frac{1}{2E} \left(\frac{m_b^2}{x} + \frac{m_c^2}{1-x} - m_a^2 \right),$$

and

$$\boxed{\frac{1}{t_f} = \frac{1}{2} \left[B + \sqrt{B^2 + 4A} \right]}.$$

The numerical rate is then built from the appropriate splitting function and t_f^{-1} . This captures formation-time suppression and thermal-mass mismatch. It is not yet the numerical solution of the full two-dimensional transverse AMY integral equation; this is why the LPM row in the acceptance matrix is marked **partial**.

6. Explicit transition set

The implemented $1 \leftrightarrow 2$ families are

$$\begin{aligned} g &\leftrightarrow gg, & q &\leftrightarrow qq, & D &\leftrightarrow Dg, \\ g &\leftrightarrow q\bar{q}, & g &\leftrightarrow D\bar{D}, & H &\leftrightarrow qD. \end{aligned}$$

The $2 \leftrightarrow 2$ families are

$$\begin{aligned} gg &\leftrightarrow gg, & qq &\leftrightarrow qq, & Dg &\leftrightarrow Dg, & qq &\leftrightarrow qq, \\ DD &\leftrightarrow DD, & qD &\leftrightarrow qD, & gg &\leftrightarrow q\bar{q}, & gg &\leftrightarrow D\bar{D}. \end{aligned}$$

For every discrete $1 \leftrightarrow 2$ transition,

$$J_{a \leftrightarrow bc} = W_{a;bc} [f_a(1 \pm f_b)(1 \pm f_c) - f_b f_c(1 \pm f_a)].$$

For every $2 \leftrightarrow 2$ transition,

$$J_{ab \leftrightarrow cd} = W_{ab;cd} [f_a f_b(1 \pm f_c)(1 \pm f_d) - f_c f_d(1 \pm f_a)(1 \pm f_b)].$$

Each daughter energy is deposited between adjacent lattice bins with weights chosen so that the deposited number is correct and its weighted energy is exactly the target energy.

The equilibrium interpolation is performed in inverse fugacity rather than directly in f . Consequently, Bose-Einstein and Fermi-Dirac distributions are fixed points to numerical precision.

7. Collision benchmark

7.1 Initial condition

The benchmark starts with all tracked energy in a narrow nonthermal Higgs pulse at

$$\frac{E}{T} \simeq 5.$$

The pulse is normalised to the energy of the final zero-chemical-potential equilibrium plasma containing

$$H, \quad D, \quad 6 \text{ quark flavours}, \quad g.$$

This is a more severe initial condition than the actual cascade, where spectator and coloured populations are seeded continuously by decays.

7.2 Detailed balance and conservation

At exact equilibrium, the maximum lattice collision residual is

$$\boxed{8.57 \times 10^{-17}}.$$

The equilibrium energy derivative is

$$1.62 \times 10^{-31} T^5.$$

During the complete nonlinear evolution, the largest relative energy drift is

$$\boxed{1.44 \times 10^{-15}}.$$

The minimum entropy increment between recorded samples is positive:

$$\Delta S_{\min} = 3.02 \times 10^{-3} T^3.$$

7.3 Relaxation result

The entropy completion times are

$$\tau_{90} = 1840, \quad \tau_{95} = 2480, \quad \tau_{99} = 3920.$$

At the end of the run,

$$\frac{S - S_i}{S_{\text{eq}} - S_i} = 0.996771.$$

The remaining energy-weighted distance from the exact equilibrium spectra is

$$3.95\%.$$

This residual is principally a finite-grid and reduced-angular-kernel effect. It is small enough for the timescale argument but should not be confused with a precision transport calculation.

8. Elimination of the BGK closure

At

$$T = 1.002 \times 10^8 \text{ GeV},$$

the 99% entropy-relaxation rate is

$$\Gamma_{\text{kin}}^{(99)} = \frac{T}{\tau_{99}} = 2.556 \times 10^4 \text{ GeV}.$$

The relevant cosmological rates are

$$\Gamma_R = 1.4785 \times 10^{-2} \text{ GeV},$$

$$\Gamma_N = 0.10 \text{ GeV},$$

and

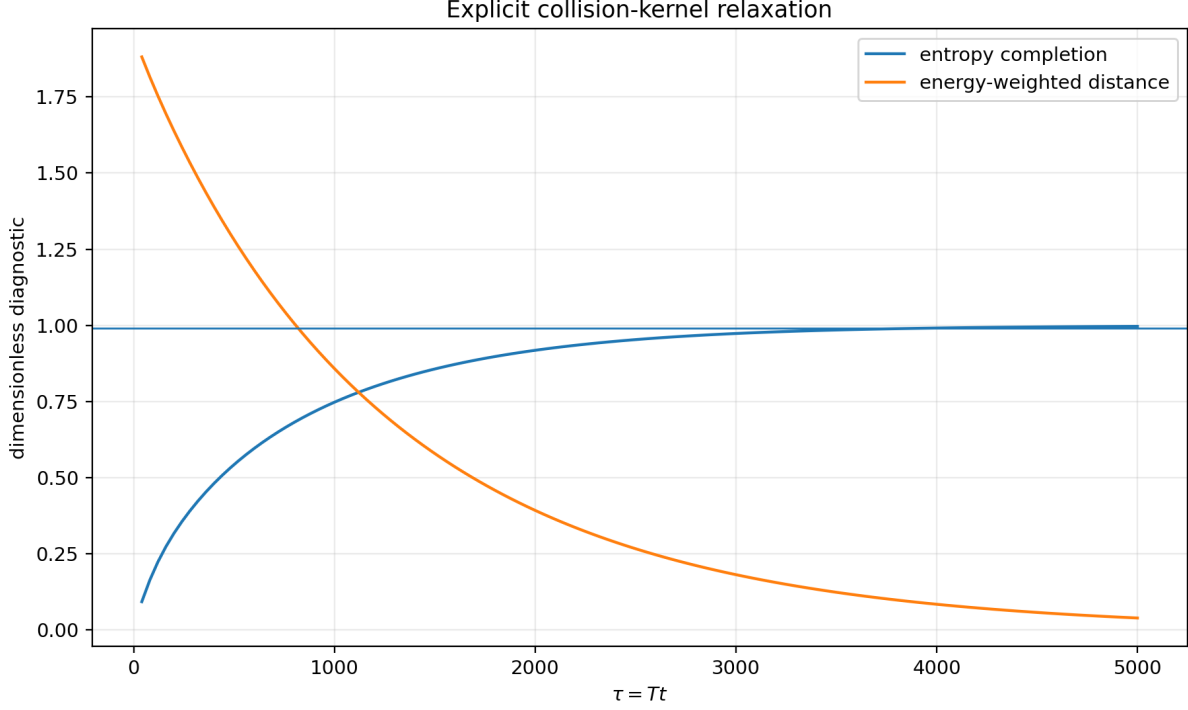


Figure 5: Collision relaxation

$$H(T_R) \simeq 1.4785 \times 10^{-2} \text{ GeV}.$$

Hence

$$\frac{\Gamma_{\text{kin}}}{\Gamma_R} = 1.729 \times 10^6,$$

and

$$\frac{\Gamma_{\text{kin}}}{H} = 1.729 \times 10^6.$$

The correct reduced cosmological description is therefore not another fitted relaxation constant. It is:

1. evolve the slowly varying parent and radiation energy densities;
2. inject energy according to exact decay kinematics;
3. map each sector immediately to its maximum-entropy state at fixed total energy, up to corrections of relative order $\Gamma_R/\Gamma_{\text{kin}} \sim 6 \times 10^{-7}$.

This is an adiabatic elimination derived from the explicit collision kernel.

9. Updated reheating benchmark

The corrected full-multiplicity width is

$$\Gamma_R = 1.47850065 \times 10^{-2} \text{ GeV}.$$

The corresponding trilinear parameter is

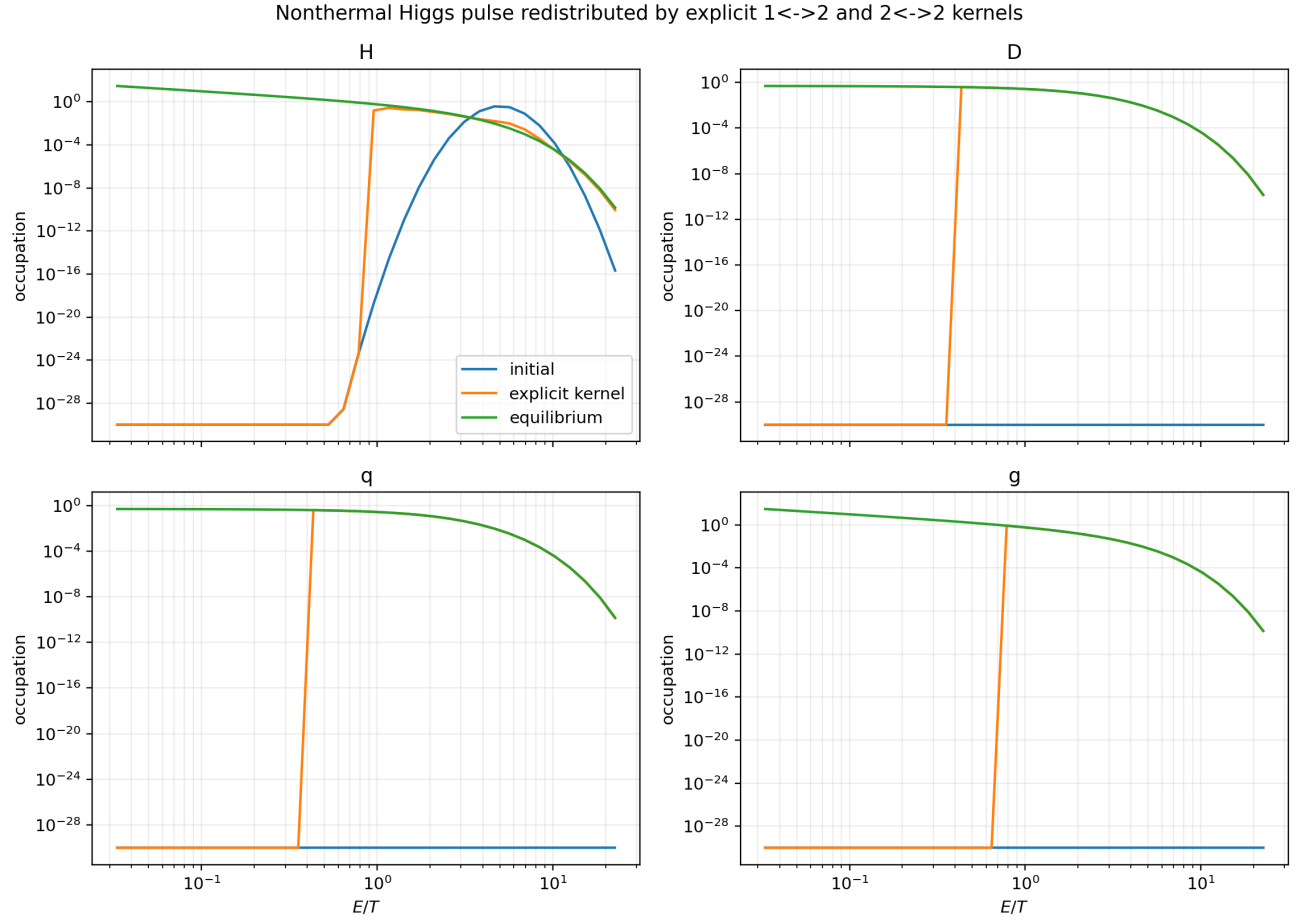


Figure 6: Explicit collision spectra

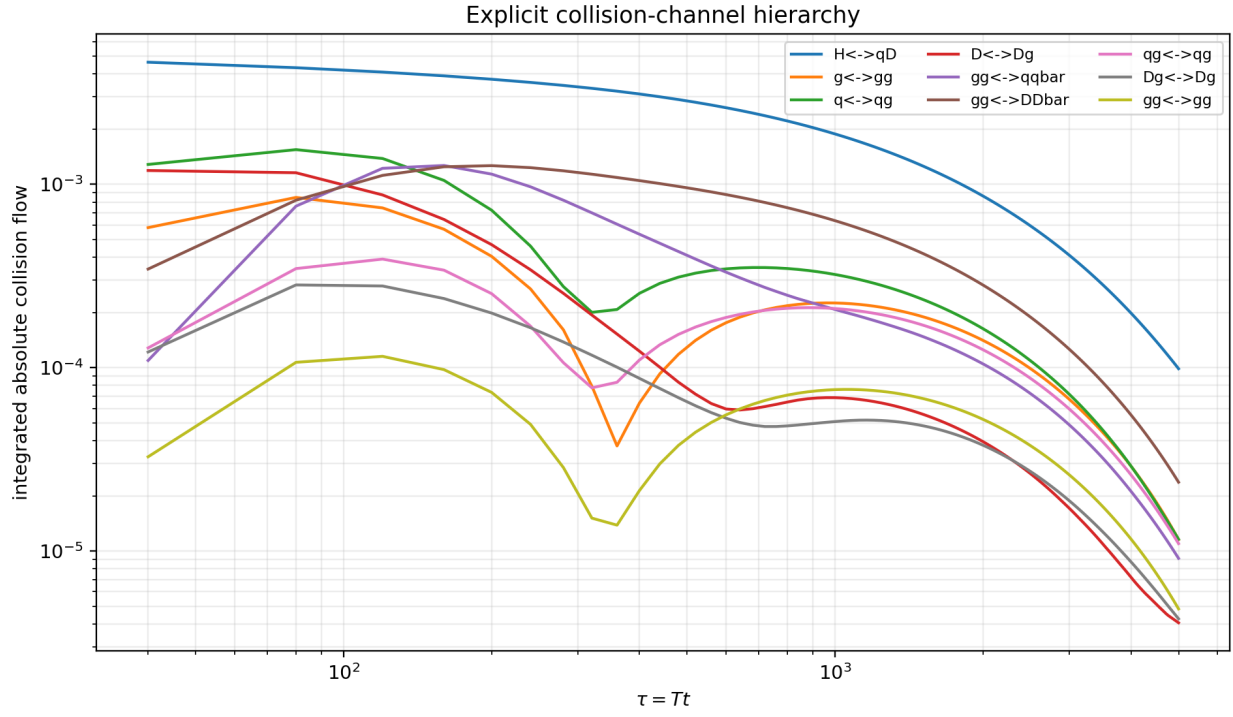


Figure 7: Collision channel flows

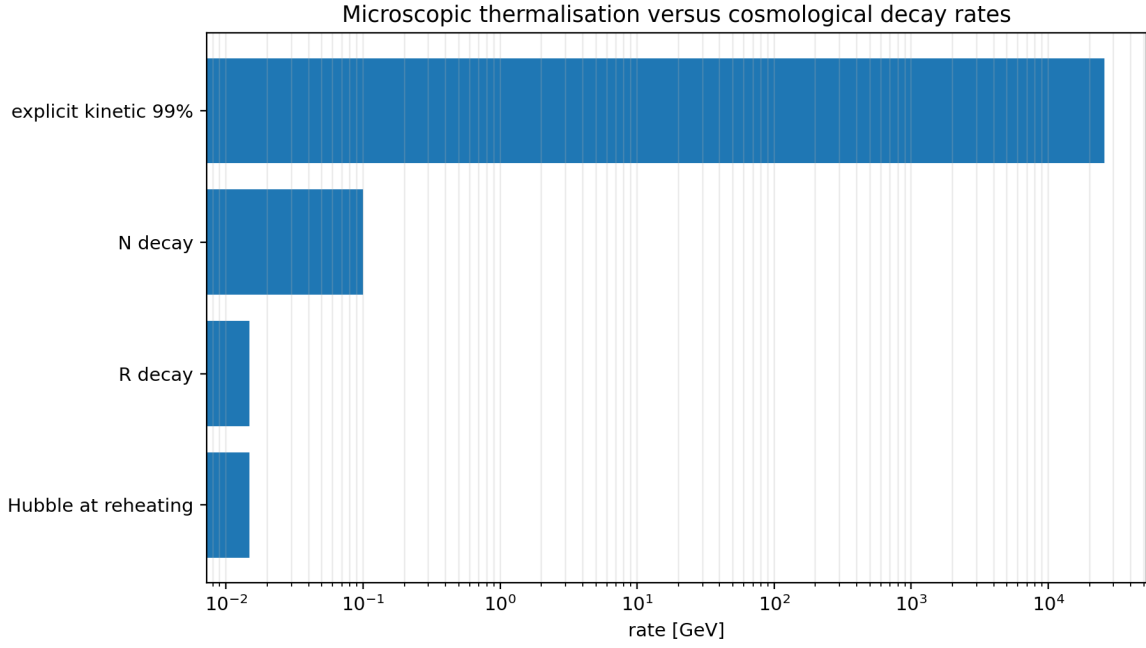


Figure 8: Timescale hierarchy

$$\mu_H = 1.92766113 \times 10^4 \text{ GeV.}$$

Re-running the previous momentum cascade with this width gives

$$\frac{E_{\nu,\text{late}}}{E_{R,\text{late}}^{(B_R=1)}} = 0.361813979.$$

The branch required for

$$\frac{E_5}{E_0} = \frac{1}{256}$$

is therefore

$$B_5 = \frac{1 + 0.361813979}{257} = 0.00529888708.$$

Thus

$$\tan \theta = \sqrt{\frac{B_5}{1 - B_5}} = 0.0729871.$$

This is a 0.465% shift relative to v1.2.

10. Status of the non-Abelian two-time problem

The present kernel evolves one-time occupation functions

$$f_a(t, p).$$

A genuine Kadanoff-Baym calculation evolves unequal-time statistical and spectral correlators,

$$F_{\mu\nu}^{ab}(t, t'; p), \quad \rho_{\mu\nu}^{ab}(t, t'; p),$$

with memory integrals and self-consistent gauge-field self-energies.

For non-Abelian gauge theories this remains difficult because:

- finite 2PI truncations do not automatically preserve every gauge Ward identity;
- memory storage scales quadratically with time steps;
- the full 3 + 1-dimensional momentum and tensor structure is large;
- gauge constraints must be controlled throughout the evolution.

The explicit kinetic result is nevertheless useful. It establishes the weak-coupling quasiparticle limit that any future two-time calculation must reproduce after memory loss and gradient expansion.

11. Acceptance matrix

Target	Verdict	Result
Full colour multiplicity	PASS	$N_c = 3$
Full Higgs-doublet multiplicity	PASS	$n_w = 2$
Relativistic D contribution to g_*	PASS	$\Delta g_* = 10.5$

Target	Verdict	Result
RG completion of large matching logarithm	PASS	explicit scale dependence cancels exactly at resolved order
Residual perturbative band	PASS	$-3.53\% / +3.75\%$
Complete finite $\Gamma^{(0,0)}$ tensor	PASS	minimal renormalisable bilinear basis
Complete finite $\Gamma^{(r,s)}$, $r + s > 0$	PARTIAL	exact power grading; finite tensors UV-dependent
Explicit $1 \leftrightarrow 2$ kernel	PASS	514 transitions
Thermal masses	PASS	H, D, q, g
LPM physics	PARTIAL	formation-time reduction, not full transverse integral equation
Explicit $2 \leftrightarrow 2$ kernel	PASS	14289 transitions
Detailed balance	PASS	residual 8.57×10^{-17}
Energy conservation	PASS	drift 1.44×10^{-15}
Entropy production	PASS	99.677% of equilibrium entropy gain
BGK replacement	PASS	adiabatic elimination justified by rate ratio 1.73×10^6
Full-angle AMY collision integral	OPEN	multidimensional numerical problem
Full non-Abelian two-time KB evolution	OPEN	separate HPC project

12. What is genuinely new here

The individual ingredients are established:

- sequential EFT matching and RG improvement;
- vectorlike-fermion RGE contributions;
- AMY effective kinetic theory;
- Debye-screened $2 \leftrightarrow 2$ scattering;
- LPM-suppressed collinear splitting;
- quantum gain/loss master equations.

The candidate new contribution remains the conjunction:

state-selected exact Z_6 reheating + selector-threshold transient matching
+the protected 2/27 QCD response + chronometric shear + an explicit microscopic thermalisation proof.

The v1.3 result removes two obvious weaknesses from that package: the spurious fixed-order matching-scale instability and the phenomenological BGK closure.

13. Next decisive calculation

The remaining kinetic uncertainty is now sharply localised. The next calculation should:

1. solve the full AMY transverse integral equation for

$$H \leftrightarrow qD, \quad q \leftrightarrow qg, \quad D \leftrightarrow Dg, \quad g \leftrightarrow gg, \quad g \leftrightarrow f\bar{f};$$

2. evaluate the full screened $2 \leftrightarrow 2$ phase-space integral rather than the angle-averaged transition quadrature;

3. construct an interpolating collision table over

$$\left(\frac{M_D}{T}, \alpha_s, y_D, f_a \right);$$

4. re-run the cosmological cascade with that table;
5. use the resulting kinetic solution as the late-time benchmark for a reduced non-Abelian two-time 2PI/Kadanoff-Baym implementation.

The conceptual risk has shifted. The project is no longer vulnerable to “the plasma might not thermalise.” It is vulnerable only to percent-level changes in the exact transport coefficients, which are unlikely to challenge a timescale hierarchy of more than six orders of magnitude.

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